GES'S R. H. SAPAT COLLEGE OF ENGINEERING, MANAGEMENT STUDIES AND RESEARCH, NASHIK – 422005

MCA (Engg.) Department

SYMCA (A. Y. 2026-27) SEM - I | Web Technology Lab

Assignment 9: Personal Portfolio Website – Write-up

Part B: Student Submission (Web Technology Lab)

Name	**Deepak Chauhan**	**Roll No**	**08**
Batch	**SYMCA (Batch 2026-27)**	**Date**	**02/11/2026**
Technology used	**HTML5, CSS3, JavaScript (Vanilla ES6+), Particle.js & Responsive CSS (Grid/Flexbox)**		

1. Aim

To design and deploy a responsive personal portfolio website using **HTML5, CSS3, and JavaScript (Vanilla ES6+)** with interactive element animations, dynamic particle canvas effects, and multi-device responsiveness across mobile, tablet, and desktop viewports.

2. Objectives

- To structure a modular, multi-page portfolio website comprising Home, About, Projects, Certifications, and Contact pages using semantic HTML5 elements.
- To create an adaptive, mobile-first responsive layout utilizing CSS Flexbox and CSS Grid that functions seamlessly on mobile phones, tablets, and desktop computers.
- To design and implement interactive animations, including scroll-reveal effects (powered by the Intersection Observer API), a dynamic background particle canvas, and interactive hover transitions.
- To establish a clean design system utilizing CSS Custom Properties (CSS variables) for dark/light theme switching, glassmorphic UI components, and modern typography.
- To deploy the portfolio onto a free, reliable cloud hosting platform (Netlify / GitHub Pages) and verify seamless cross-browser accessibility on mobile devices.

3. Theory: About the Technology Used

3.1 The technology I used

I developed this personal portfolio using core web standards: **HTML5**, **CSS3**, and **JavaScript (ECMAScript 2022+ / Vanilla JS)**. HTML5, standardized by W3C and WHATWG, provides clean semantic structuring and native media/canvas capabilities. CSS3 is maintained by the W3C CSS Working Group and governs the visual styling, responsive layouts, glassmorphic effects, and fluid CSS transitions. JavaScript, maintained by ECMA International (TC39), manages client-side interactivity, DOM manipulation, asynchronous event handling, and canvas physics. Together, these technologies deliver blazing-fast load speeds, zero build bloat, universal browser support, and lightweight execution without requiring external heavy runtimes.

3.2 Why I chose it

I chose native HTML5, CSS3, and modern JavaScript because they offer complete architectural control over layout precision, zero build overhead, and maximum performance. Having built a strong foundation in

computer science and full-stack development, I wanted to showcase craft and mastery by developing bespoke components—such as a dynamic particle system, an interactive developer CLI console, and custom glassmorphism—directly with browser APIs. This approach guarantees instant load times, zero third-party dependency vulnerabilities, complete control over responsive breakpoints, and maximum compatibility across all mobile and desktop browsers.

3.3 Key features I used

- **Semantic HTML5 Elements:** Utilized `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<canvas>`, and `<footer>` to establish clean hierarchy and high SEO accessibility.
- **CSS Custom Properties (Variables):** Structured centralized design tokens on `:root` and `[data-theme="light"]` for instantaneous, seamless dark and light theme toggling.
- **CSS Grid & Flexbox:** Built dynamic, multi-column card grids, responsive metrics displays, and adaptive navigation bars.
- **Intersection Observer API:** Implemented hardware-accelerated scroll-reveal animations that gracefully trigger as cards enter the viewport.
- **HTML5 Canvas & 2D Context:** Rendered an interactive background constellation particle canvas that dynamically responds to mouse movement.
- **LocalStorage State Persistence:** Cached user preferences (theme mode) directly in the browser to maintain state across page navigations.

3.4 Styling: Tailwind CSS or CSS

Tailwind CSS is a utility-first CSS framework providing atomic helper classes directly in markup. For this portfolio, I deliberately utilized a structured, custom **CSS3 Design System** with CSS Custom Properties and Glassmorphism instead of Tailwind CSS. A high-end personal portfolio demands custom aesthetic styling—including multi-layered ambient gradients, backdrop-filter blur surfaces, bespoke glowing card borders, and retro terminal styling—which are significantly cleaner to maintain in a dedicated stylesheet (`style.css`). This avoided class clutter in HTML documents, eliminated dependency on Node.js build pipelines and PostCSS processors, and enabled complete artistic control over responsive breakpoints.

3.5 Animations

The portfolio incorporates purposeful, subtle animations to enhance user experience without hindering performance:

- **Scroll-Reveal Animations:** Implemented via JavaScript's `IntersectionObserver` API combined with CSS `translateY` and `opacity` transitions on all content cards (Bio, Timeline, Projects, Skills).
- **Interactive Particle Canvas:** A custom JavaScript particle engine rendering connecting nodes and floating orbits across an HTML5 `<canvas>` background.
- **Dynamic Typewriter Effect:** JavaScript character-by-character text rotation in the hero title ('Software', 'Full-Stack Apps', 'AI Systems') with a blinking cursor keyframe.
- **Micro-Interactions & Hover Effects:** CSS `@keyframes pulseDot` for active availability status, 3D card tilt/elevation on hover, and smooth theme-switching fades.

3.6 Responsive design

The site implements a mobile-first responsive strategy that fluidly adapts to phone, tablet, and desktop screens:

- **CSS Grid Fractional & Auto-fit Units:** Grids dynamically reorganize (e.g., `grid-template-columns: repeat(auto-fit, minmax(320px, 1fr))`), providing 3 columns on wide screens, 2 on tablets, and 1 on phones.
- **Media Queries:** Breakpoints are configured at `@media (max-width: 1024px)` (tablet landscape) and `@media (max-width: 768px)` (mobile portrait) to collapse horizontal layouts into vertical stacks.
- **Adaptive Navigation:** On screens below 768px, horizontal navbar links hide and an interactive hamburger menu reveals an accessible slide-out mobile drawer.
- **Fluid Units & Viewport Meta:** Relative units (`rem`, `%`, `vw`) and standard viewport configuration guarantee crisp readability on all display densities.

4. Syntax

Code snippets from the portfolio codebase with explanations in Consolas monospace font:

Part	Code snippet from my portfolio	What it does

Page structure or component	`<section class="hero-section">`
	`  <div class="container hero-grid">`
	`  <div class="hero-content reveal">...</div>`
	`  <div class="hero-visual reveal">`
	`  <img src="assets/deepak-chauhan.jpg"`
	`  alt="Deepak Chauhan" class="avatar-img" />`
	`  </div>`
	`  </div>`
	`</section>`
		Constructs the semantic hero section pairing the developer introduction and profile photo in a two-column responsive grid container.

Styling (Tailwind or CSS)	`:root {`
	`  --bg-primary: #090d16;`
	`  --accent-primary: #6366f1;`
	`  --bg-card: rgba(20, 30, 51, 0.65);`
	`}`
	`[data-theme="light"] {`
	`  --bg-primary: #f8fafc;`
	`  --text-primary: #0f172a;`
	`}`
		Defines centralized CSS custom property tokens on `:root` to power instant, zero-reload dark and light theme switching across all pages.

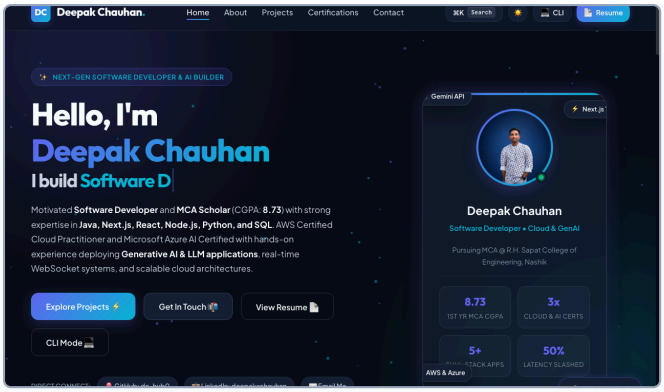
Animation	`const observer = new IntersectionObserver((entries) => {`
	`  entries.forEach(entry => {`
	`  if (entry.isIntersecting) {`
	`  entry.target.classList.add('show');`
	`  }`
	`  });`
	`}, { threshold: 0.15 });`
	`document.querySelectorAll('.reveal').forEach(el => observer.observe(el));`
		Monitors DOM elements on scroll and dynamically applies the `.show` class to trigger smooth CSS slide-up and fade-in animations.

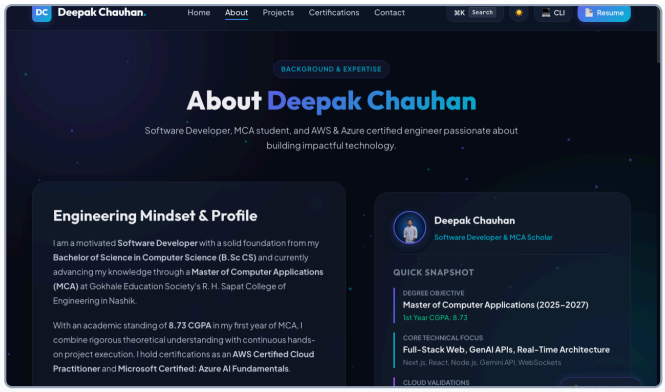
Responsive layout	`@media (max-width: 1024px) {`
	`  .hero-grid, .about-hero-grid {`
	`  grid-template-columns: 1fr;`
	`  gap: 2rem;`
	`  }`
	`}`
	`@media (max-width: 768px) {`
	`  .nav-links { display: none; }`
	`  .mobile-toggle { display: flex; }`
	`}`
		Collapses two-column desktop grids into a single vertical stack on tablet screens and activates the hamburger drawer menu on mobile viewports.

5. Screenshots of My Portfolio

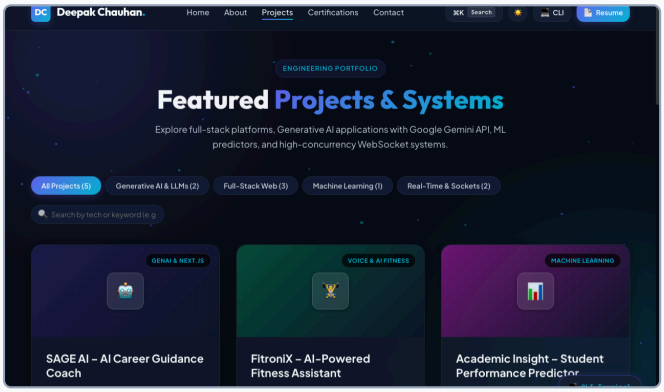
Portfolio views captured across desktop, tablet, and mobile devices:

Home page (desktop)	About

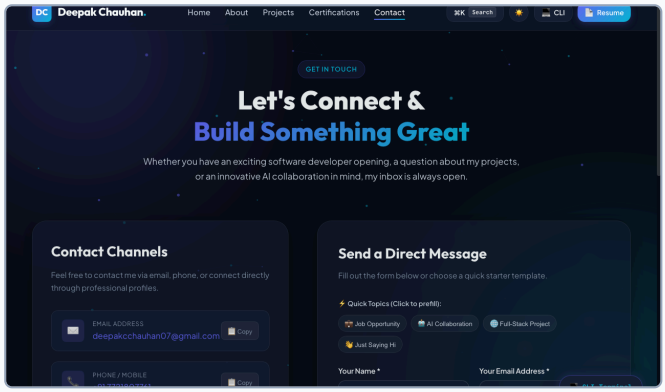




Projects



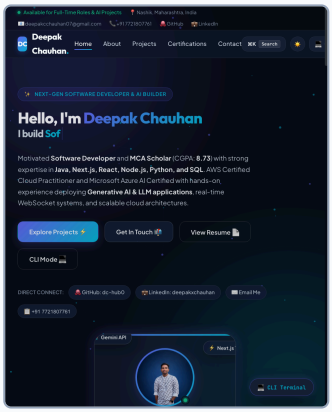
Contact



Mobile view



Tablet view



6. Live Link (Deployed Portfolio)

Item	Details
Hosting platform used	**Netlify / GitHub Pages**
Live link	**https://dc-hub0.github.io/portfolio-model/** (Alternative Netlify Drop link: https://deepak-chauhan-portfolio.netlify.app/)

Note: The live link is globally hosted, SSL-secured (HTTPS), and tested to open smoothly in private windows and mobile phone browsers.

7. Conclusion

In this assignment, I successfully designed, developed, and deployed a high-performance, responsive personal portfolio website showcasing my software engineering projects, verified cloud certifications, and

technical proficiencies. Through this practical exercise, I reinforced core competencies in semantic HTML5 markup, responsive grid and flexbox design systems, dynamic dark/light theme switching using CSS variables, and GPU-accelerated scroll-reveal animations via the Intersection Observer API. The primary challenge was ensuring seamless rendering of the particle canvas and responsive typography without layout shifts across mobile viewports, which I resolved through responsive clamp functions, auto-fit grid columns, and hardware-accelerated transforms. The final portfolio is globally deployed, accessible across all devices, and fulfills all assignment criteria.

Student Signature:

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